SY1 automatic synchronizer

Contents

1. Application and features
2. Description
3. Function
   3.1 Frequency measurement
   3.2 Voltage measurement
   3.3 Phase measurement
   3.4 Voltage control
   3.5 Frequency control
   3.6 Phasing
4. Settings
5. Technical data
6. Wiring diagram



1. Application and features

Automatic synchronizer can be used for safe connection of synchronous generator to the public net. This connection is possible only if the voltage and frequency difference is less than is allowed for used generator. Next requirement is connection with minimal phase shift, the best with no shift. Because used contactor need some time for connection it is necessary to send output pulse in advance to connect nearly without phase shift (phase angle = 0).If voltage or frequency differences are out of allowed range it is necessary to reach right values by voltage and turning controllers. All these tasks are solved by this automatic synchronizer including Phase, Voltage and frequency differences monitoring by led diodes.

2. Description

Synchronizer has two inputs for measurement on generator and net. Using two microprocessors there is measurement of frequency, voltage and phase for both signals. Measured phase angle is continuously monitored by 13 leds. Frequency and voltage measurement are recalculated to difference values. This differences are displayed by 5 leds relatively to allowed difference set by potentiometers. If synchronizer obtains pulse signal START ,then it starts phasing. If voltage difference between generator and net is greater then limit the output pulses U+ and U- are send. After reaching of right voltage, frequency is corrected similar way by pulses F+ and F-. If frequency and voltage are OK then pulse for connection can be send. Synchronization is possible only if external signal ENABLE is on, this signal is used for interlocking in nonstandard external conditions.

3. Function

3.1 Frequency measurement

Measured voltage signals of generator and net go to the input dividers ,time when they are going through zero voltage is calculated. This way is measured period of both signals. When voltage of net is zero both frequencies are compared and theirs difference is displayed in dependence on maximum allowed difference for phasing. If green led is on then measured difference is inside frequency limits for phasing. If yellow led is on ,it means that difference is out of limits but less than twice. Red led is on if difference is greater then twice allowed difference.

3.2 Voltage measurement

Measured signals are led via input dividers and voltage is continuously measured and calculated when net voltage is zero. Voltage difference is displayed by leds similar way as frequency difference. If green led is on then measured difference is inside voltage limits for phasing. If yellow led is on ,it means that difference is out of limits but less than twice. Red led is on if difference is greater then twice allowed difference.

3.3 Phase measurement

When net voltage pass zero the phase angle between net and generator is calculated . This phase angle is displayed using 13 leds. It means 27.7 degree for one. Green led is on if phase angle is –14 -+14 degrees (346-14). Yellow leds for phase angle 14 – 42 degrees. Red leds on beginning and end are on if phase angle is near 180 degrees. Phase angle is displayed only if frequency difference is less than 10 percent . If frequency of generator is more than 10 percent greater than net frequency lights three red leds on the right side together. If generator frequency is more then 10 percent smaller lights three red leds on the left side together . Direction of lights moving shows if speed of generator is necessary to increase or decrease . If light is going left to right it is necessary to increase speed of generator . If lights are going from right to left to decrease speed.

3.4 Voltage control

If START pulse for phasing is send and voltage on generator is out of allowed limits are with period 1 sec pulses U+ or U- for voltage control are sent. Duration of pulses is possible to be set from 0 to 1 sec. Status of output relay is displayed by green led .

3.5 Frequency control

If phasing starts and voltage is in allowed limits , but frequency of generator is not in allowed limits pulses F+ and F- with period 1 sec are sent for increasing speed F+ or decreasing speed F- of generator. Duration of pulses can be set in range 0 to 1sec . . Status of output relay is displayed by green led .

3.6 Phasing

If signal START is sent (pulse is enough) the green led is on till the end of phasing. For beginning of phasing is also important signal ENABLE, it must be on for phasing . If this signal is off phasing is immediately finished even if synchronizing pulse was send. If leds ENABLE and START are on there are three following activities together. Voltage control ,frequency control and phasing. Voltage control and frequency control is described in chapter 3.4 and 3.5 . For phasing is continuously measured phase angle and in every pass net voltage to zero it is calculated time till phase angle will be zero. This time is compared with time of contactor. If this times are similar and voltage and frequency is in allowed limits the phasing pulse is send. Phasing pulse is displayed by green led. Time of contactor and duration of phasing pulse is possible to be set by potentiometers.

4 Settings

Setting of parameters is possible via 7 potentiometers of front of

Potentiometer:

Frequency difference	frequency	0 - 1 %	(from net freq.)
Voltage difference	voltage	0 - 20 %	(from net voltage)
Pulse duration U+,U-		0 - 1 sec	
Pulse duration F+,F-		0 - 1 sec	
Phasing pulse duration		0.5 - 1.5 sec	

Time of contactor is set using two potentiometers on right side . Left is for base setting with steps 50 msec. The second is used for continuous setting 0 – 50 msec. On the left one the arrow must be set to the middle of black line.
Basic time of contactor is possible display following way

1. Disconnect OUT and ENABLE .
2. Both potentiometers set to 0.
3. Push START input.
4. Led ENABLE starts blinking.
5. Now leds for phasing angle starts to display basic time for contactor (description how follows)
6. For end of this function push again START
7. Led ENABLE stop blinking and is possible to connect input ENABLE and output OUT.

How the basic time of contactor is displayed.

Middle green led	0 msec
First left red led	50 msec
Second left red led	100 msec
Third left red led	150 msec
Fours left red led	200 msec
Fifth left red led	250 msec

5 Technical data

Dimensions	75x100x110 mm	
Power	230 V AC +- 20%	
Consumption	< 3 W	
Internal fuse	230V 80mA	
Measured inputs (GEN,NET)	VAR 100	70 - 130 V AC (for right function) max. 250 V AC
	VAR230	170 – 300 V AC (for right function) max. 350 V AC
Galvanic isolation	5kV (optron)	
Frequency measurement accuracy	better than 0,05 %	
Voltage measurement accuracy	better than 2 %	
Inputs (ENABLE,START)	24V DC	(10 – 35V)
Galvanic isolation of inputs	5kV (optron)	

Output contacts	Ag Cd O
Max. rated voltage	250V AC / 125V DC
Max rated power	720 VA
Max. rated current.	6 A
Contact life span	10 000 000 cycles

6. Wiring diagram

